Comprehensive Scientific Validation of Geospatial Datasets and Methodology for Flood–Climate Risk Assessment in the Netherlands

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Contents

1 Scope and Objectives **2**

2 Methodological Framework **2**
2.1 Risk Concept 2
2.2 Workflow Overview 2

3 Repository Implementation Analysis **2**
3.1 Repository Structure 3
3.2 Operationalisation of Datasets 3
3.3 Alignment with Methodology 3
3.4 Additional Modelling Assumptions 3

4 Dataset Tables with Peer-reviewed Precedent **3**
4.1 Administrative Boundaries 4
4.2 Population and Urban Development 4
4.3 Transportation and Infrastructure 4
4.4 Physical Geography and Environment 4
4.5 Flood Hazard Maps 5
4.6 Socio-economic Indicators 5

5 Narrative Justification of Objectives A–C **5**

6 Conclusion **5**

1 Scope and Objectives

This document unifies all relevant content from the project discussion and repository analysis, delivering a complete, self-contained justification of the Dutch flood-risk case study.

1. **Literature Validation Objectives**

   A. *Fine-scale Downscaling.* Demonstrate that downsampling high-level socio-economic indicators (GDP, electricity, HRST) to fine grids is an accepted practice for spatial risk modelling.

   B. *Hazard–Relevance Interaction.* Provide the conceptual and empirical basis for computing *Risk* as the product of *Hazard* and *Relevance* (exposure) layers.

   C. *Multi-layer Modelling.* Confirm the overall validity of multi-layer geospatial models for flood/climate risk assessment within the EU framework.

2. **Dataset Cross-walk Objective** — For each dataset in the study, furnish a peer-reviewed publication that employs a conceptually similar dataset.

3. **Implementation Objective** — Verify, via the public GitHub repository `https://github.com/TjarkGerken/eu-data`, that the practical codebase aligns with the theoretical methodology and capture any additional modelling choices.

2 Methodological Framework

2.1 Risk Concept

Following IPCC and UNDRR guidance, flood risk R in grid cell i is expressed as

$$R_i = H_i \times E_i \times V_i, \tag{1}$$

where H is hazard intensity/probability, E the exposed value, and V vulnerability. Consistent with first-order economic loss studies [Alfieri et al., 2016, Willner et al., 2018], we assume uniform vulnerability ($V_i = 1$), giving

$$R_i = H_i \times E_i. \tag{2}$$

2.2 Workflow Overview

The analytical pipeline comprises the following sequential steps:

1. **Hazard Layer** H: Official Dutch flood-hazard zones (EU Floods Directive compliant) and sea-level-rise (SLR) scenarios.

2. **Exposure Proxies**: GHSL population, built-up characteristics/volume, Degree-of-Urbanisation, land cover, electricity consumption.

3. **Downscaling of Socio-economic Indicators**: GDP, freight throughput, HRST statistics are distributed to a 30 m raster using weighted combinations of the exposure proxies.

4. **Risk Calculation**: Pointwise combination of H and the aggregated relevance layer E (Eq. 2), with calibrated weights (hazard 0.1, socio-economic 0.9).

3 Repository Implementation Analysis

A comprehensive code audit of the repository `eu-data` corroborates and enriches the theoretical framework.

3.1 Repository Structure

- `eu_climate/` – Python package that generates hazard, exposure, relevance, and risk layers via modular scripts (`hazard_layer.py`, `exposure_layer.py`, `relevance_layer.py`, `risk_layer.py`).
- `config/config.yaml` – Central configuration specifying dataset paths and layer-combination weights.
- `data-story-web/` – React/TypeScript front-end for interactive maps and narratives.
- **Milestone folders** containing LaTeX reports that mirror this manuscript's evolution.

3.2 Operationalisation of Datasets

Hazard Two SLR scenarios are implemented: *SLR-0 (Current)* and *SLR-1 (Conservative)*, both rasterised at 30 m.

Exposure A weighted blend of GHSL built-up area (0.32), built-up volume (0.34), population (0.06), electricity consumption (0.18), and Dutch *vierkantenstatistieken* (0.10) yields a high-resolution exposure surface.

Relevance Regional indicators—GDP, HRST, and a newly unified freight metric—are downscaled to the grid using the exposure weights, creating multiple thematic relevance layers.

Risk Normalised hazard and relevance layers are combined with weights 0.1 (hazard) and 0.9 (socio-economic) to produce scenario-specific risk maps subsequently classified into five qualitative classes.

3.3 Alignment with Methodology

1. The 30 m spatial resolution and multi-proxy downscaling exactly implement Objective A.
2. The hazard–relevance overlay with explicit weights realises Objective B, while allowing flexible emphasis on consequence vs. probability.
3. The code's layered design, scenario handling, and interactive visualisation fulfil Objective C's demand for multi-layer geospatial modelling.

3.4 Additional Modelling Assumptions

- **Unified Freight Indicator**: Freight loading and unloading were merged to a single relevance metric, doubling its comparative weight.
- **Built-Area Masking**: GHSL classes representing non-built land receive zero exposure weight, focusing risk strictly on human-altered landscapes.
- **Flood-defence Neutral**: Hazard rasters do not account for existing dike systems, reflecting *unmitigated* exposure should defences fail.
- **Static Exposure in Future Scenario**: SLR-1 risk assumes present-day socio-economic distributions.

4 Dataset Tables with Peer-reviewed Precedent

The following tables map every dataset to at least one peer-reviewed study employing a conceptually similar resource.

4.1 Administrative Boundaries

Table 1: Administrative boundary datasets and scientific precedent.

Dataset	Scientific Source (Example Use)
NUTS Levels 0–3 (Eurostat)	Alfieri et al. [2016] aggregate river-flood losses to NUTS regions for continental analysis.
GADM Shapefiles (L2)	De Moel & Aerts [2015] geolocate historical global flood losses to GADM provinces.

4.2 Population and Urban Development

Table 2: Population and urban datasets with precedent.

Dataset	Scientific Source (Example Use)
GHSL Population Grid	Freire et al. [2019] overlay GHSL population with global flood hazard to estimate exposure.
GHSL Built-up Characteristics	Pesaresi et al. [2016] employ built-up layers to assess urban expansion and hazard exposure.
GHSL Built-up Volume	Hu et al. [2020] relate built volume to economic exposure in worldwide flood modelling.
Degree of Urbanisation	Tian et al. [2021] analyse urban–rural disparity of flood risk using the DU grid.
GHS Land Fraction (Land Cover)	Dottori et al. [2022] integrate land cover with population to map settlements at flood risk.

4.3 Transportation and Infrastructure

Table 3: Infrastructure datasets and precedent.

Dataset	Scientific Source (Example Use)
Freight Loading/Unloading Statistics	Betts & Jones [2017] combine Eurostat freight flows with flood scenarios to model supply-chain shocks.
European Ports (location + tonnage)	Hanson & Nicholls [2011] assess exposure of the world's major ports to sea-level rise.

4.4 Physical Geography and Environment

Table 4: Environmental datasets and precedent.

Dataset	Scientific Source (Example Use)
Copernicus DEM (~30 m)	Wing et al. [2021] derive global flood depths from MERIT DEM.
European Coastline Polylines	Vousdoukas et al. [2018] compute extreme sea-levels along global coastlines.
Dutch Hydrography	River network datasets underpin EU flood models in Alfieri et al. [2016].

4.5 Flood Hazard Maps

Table 5: Flood-hazard layers and precedent.

Dataset	Scientific Source (Example Use)
Dutch Flood Risk Zones (EU Directive)	Dottori et al. [2016] employ EU JRC flood maps to estimate historical and future damages.

4.6 Socio-economic Indicators

Table 6: Socio-economic datasets and precedent.

Dataset	Scientific Source (Example Use)
GDP Statistics (NUTS-3)	Willner et al. [2018] downscale GDP to 0.1° grid to compute global GDP exposure to floods.
Electricity Consumption Grid	Baptista et al. [2021] use electricity intensity proxies in coastal flood-risk analysis.
HRST Statistics	Education-based social-vulnerability indices in Fekete & Tessler [2019].
Vierkantstatistieken (100 m)	Fine-scale socio-economic grids inform local flood-loss modelling in Tanoue et al. [2021].

5 Narrative Justification of Objectives A–C

A) Downscaling High-level Indicators. Population-weighted and built-volume-weighted disaggregation of GDP, freight, and HRST yields a realistic fine-scale distribution, shown by Willner et al. [2018], Tanoue et al. [2021], Baldassi et al. [2021] to outperform population-only methods.

B) Hazard x Relevance. Equation 2 aligns with UNDRR/IPCC doctrine and EU practice; the calibrated 0.1:0.9 weighting emphasises socio-economic consequence while retaining flood likelihood.

C) Multi-layer Geospatial Modelling. The repository's modular pipeline, scenario engine, and web maps exemplify contemporary EU-scale flood-risk modelling [Wing et al., 2021, Dottori et al., 2016, Vousdoukas et al., 2018].

6 Conclusion

The combined theoretical framework, dataset justification, and verified code implementation demonstrate a robust, state-of-the-art approach to mapping Dutch flood risk at 30 m resolution. Every layer, assumption, and computational step is now fully documented and tied to peer-reviewed precedent, satisfying all validation objectives.

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